

Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Momina Asim

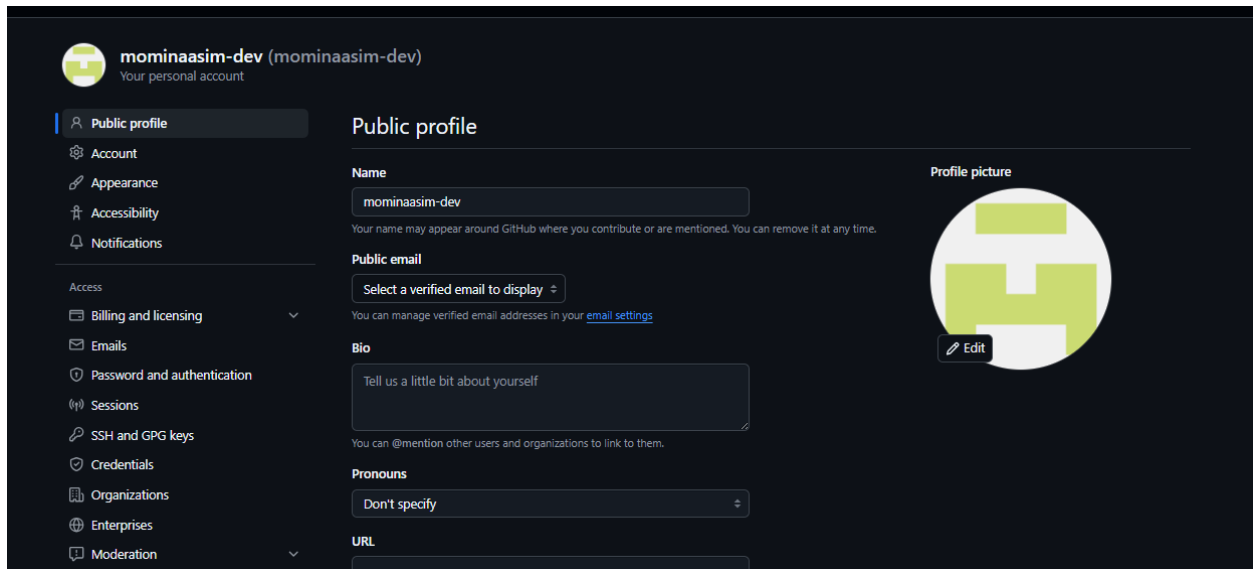
Registration number: fa24b1-se-069

Course code: Cloud Computing

Section: A

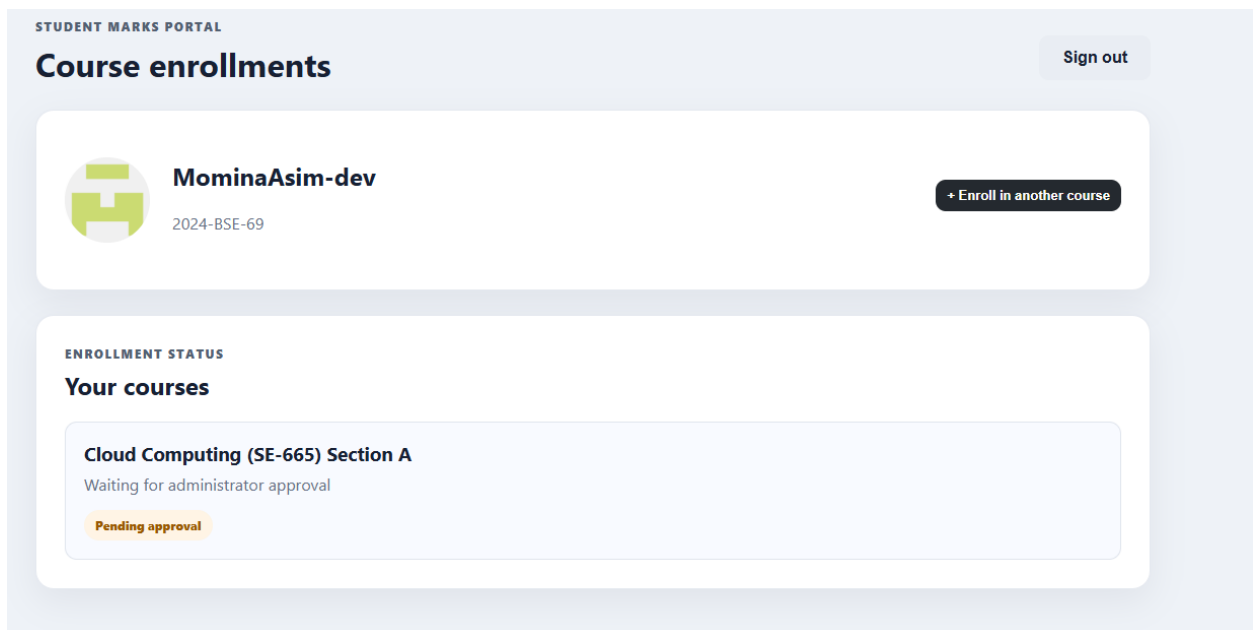
Github username: mominaasim-dev

Task 01: GitHub Account Setup



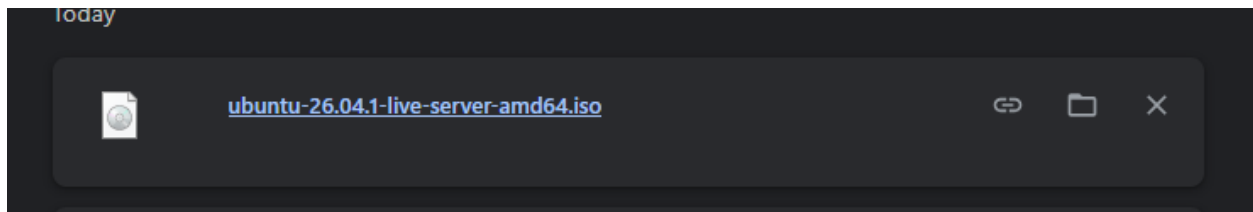
Shows my GitHub profile with my complete actual name visible.

Task 2: Student Marks Portal Registration

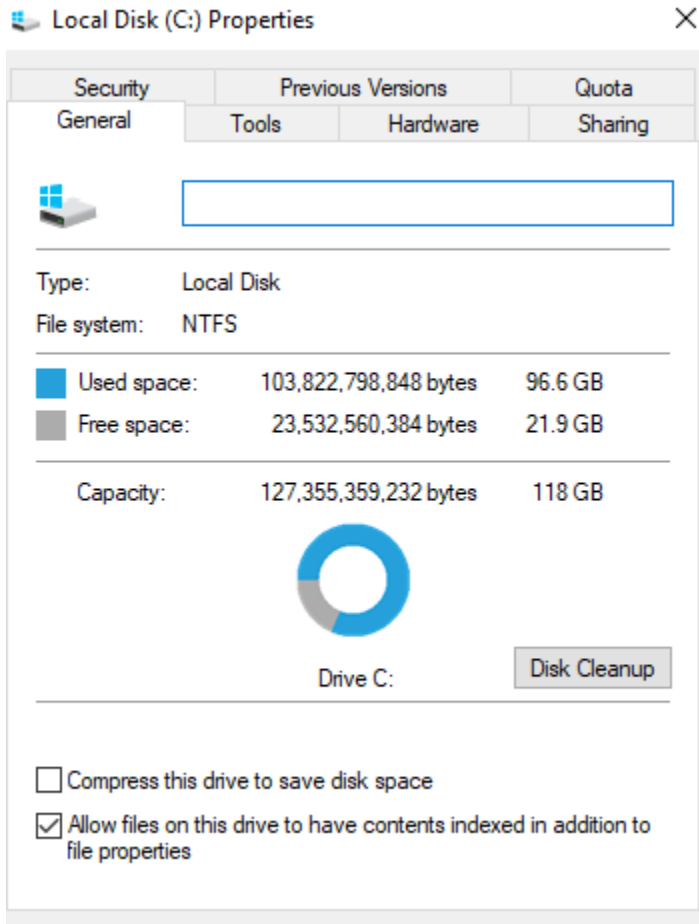


- **student_portal_registration.png** — *Shows successful registration on the Student Marks Portal using my GitHub account.*

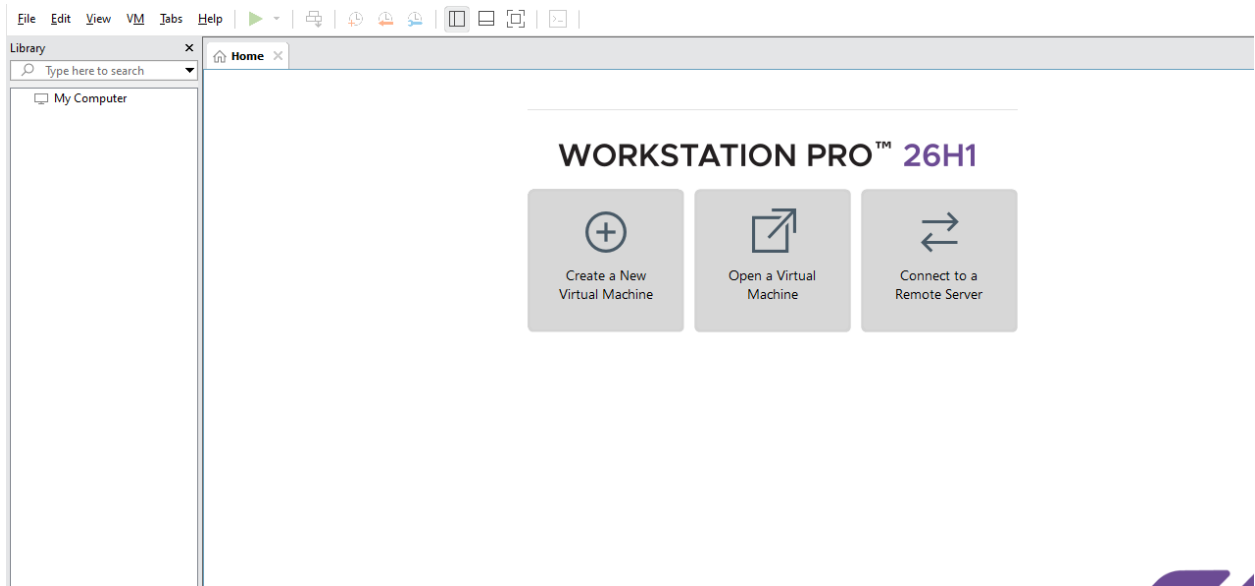
Task 3: Download Ubuntu Server ISO and Verify Storage



- **ubuntu_server_iso.png** — *Shows the downloaded Ubuntu Server 26.04.1 LTS ISO filename in the Downloads folder.*

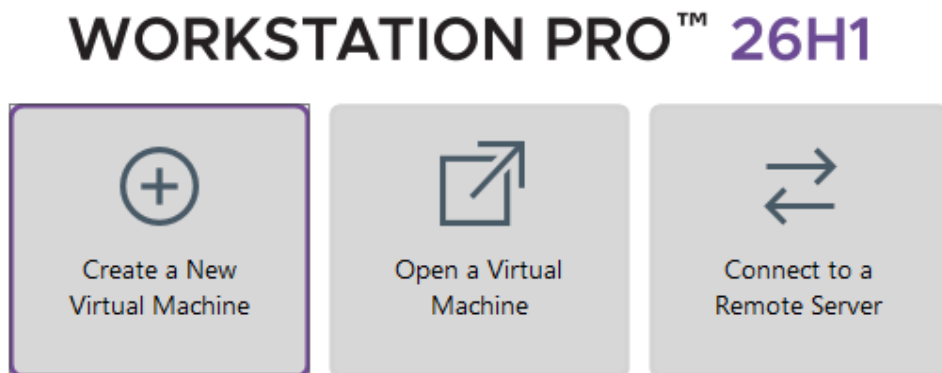


- **available_storage.png** — *Shows at least 20 GB of free storage available on Local Disk (C:) for the Ubuntu Server VM.*



VMware_work_station.png ensure installation of it.

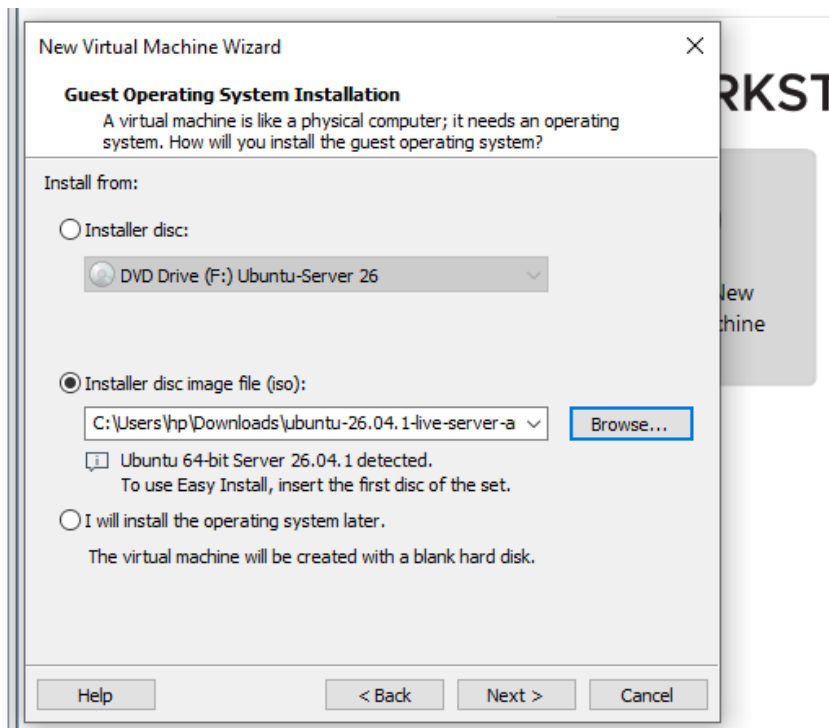
Task 4: Create the Ubuntu Server Virtual Machine



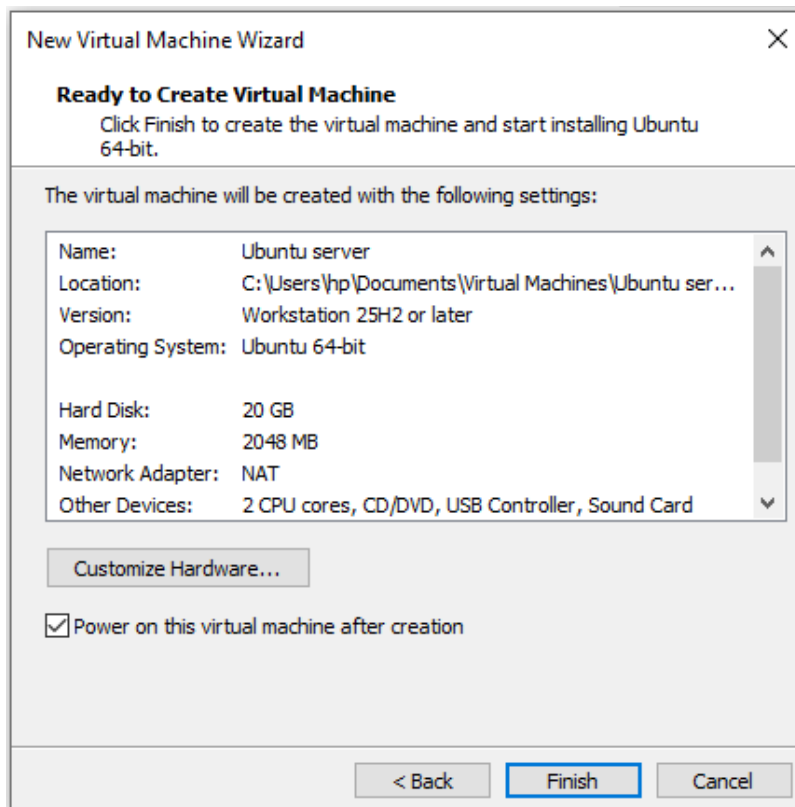
- **vm_creation_wizard.png** — *Shows the New Virtual Machine Wizard opened in VMware Workstation Pro.*



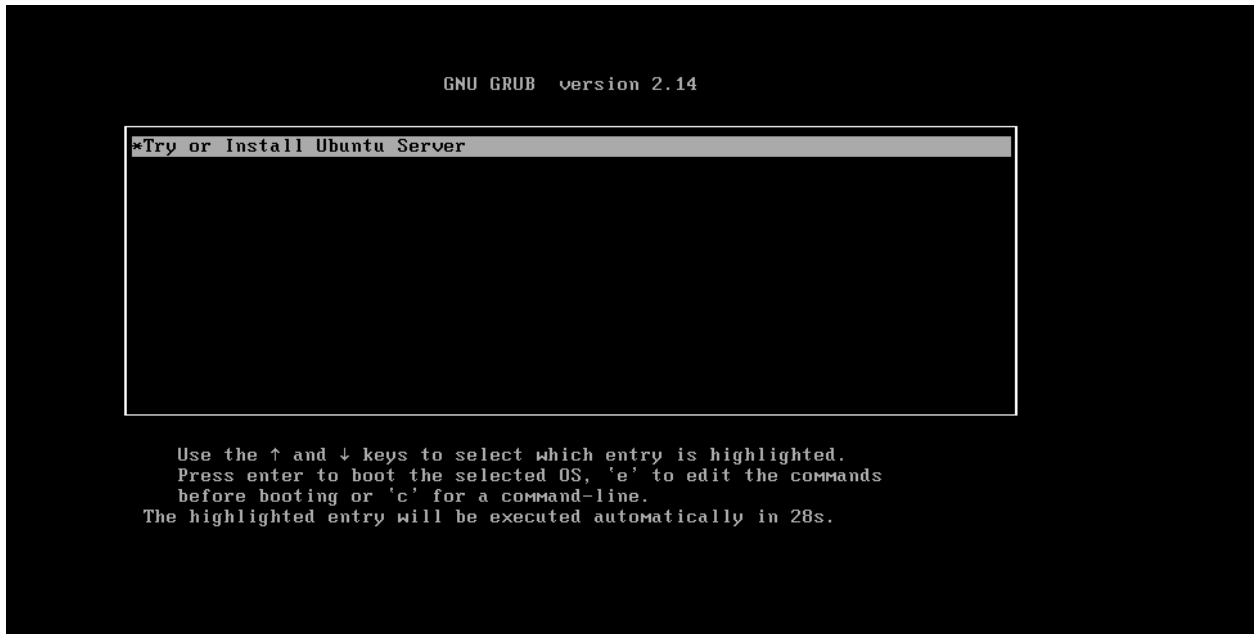
- **vm_typical_configuration.png** — *Shows “Typical (recommended)” selected as the VM configuration.*



- **vm_ubuntu_iso_selected.png** — *Shows the Ubuntu Server ISO path selected as the installer disc image.*

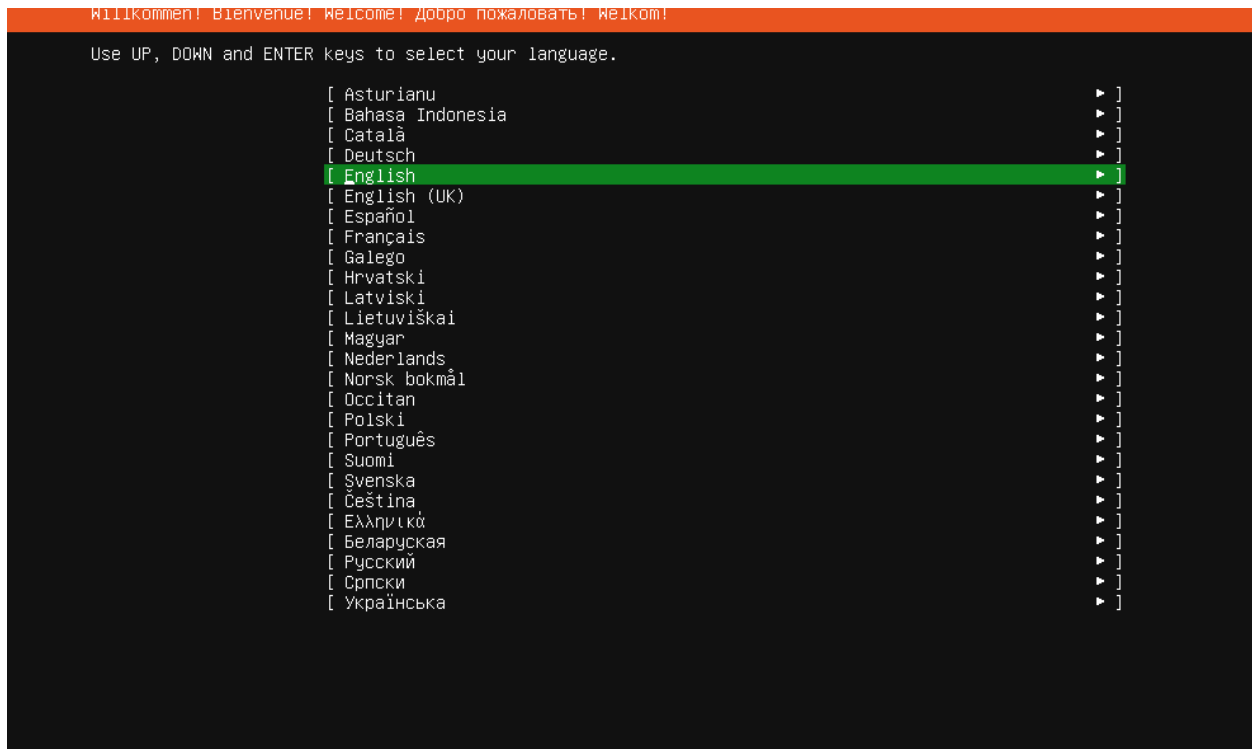


- **vm_configuration_summary.png** — *Shows the final VM configuration summary with at least 20 GB virtual disk.*



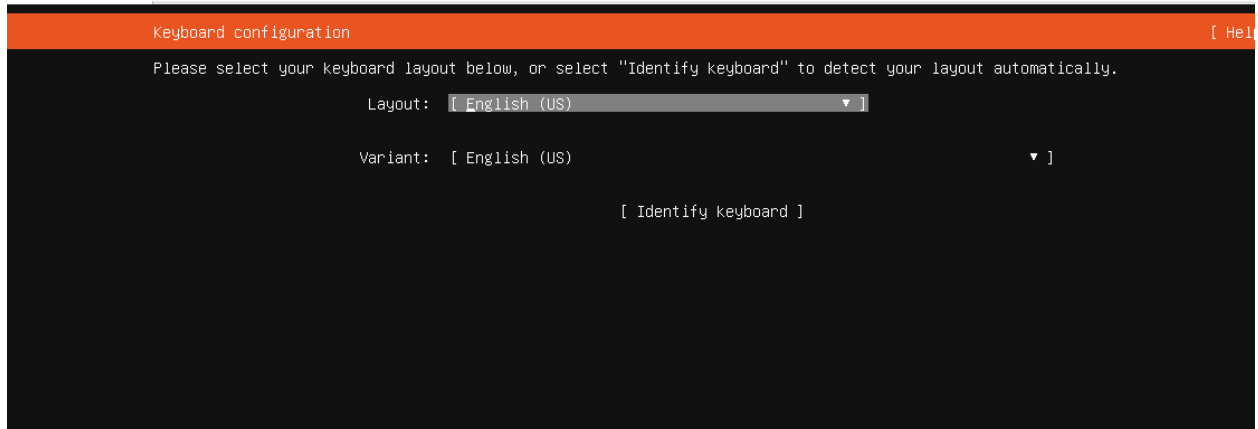
- **ubuntu_installer_boot.png** — *Shows the Ubuntu Server boot menu with “Try or Install Ubuntu Server” highlighted.*

Task 5: Choose the Installation Language

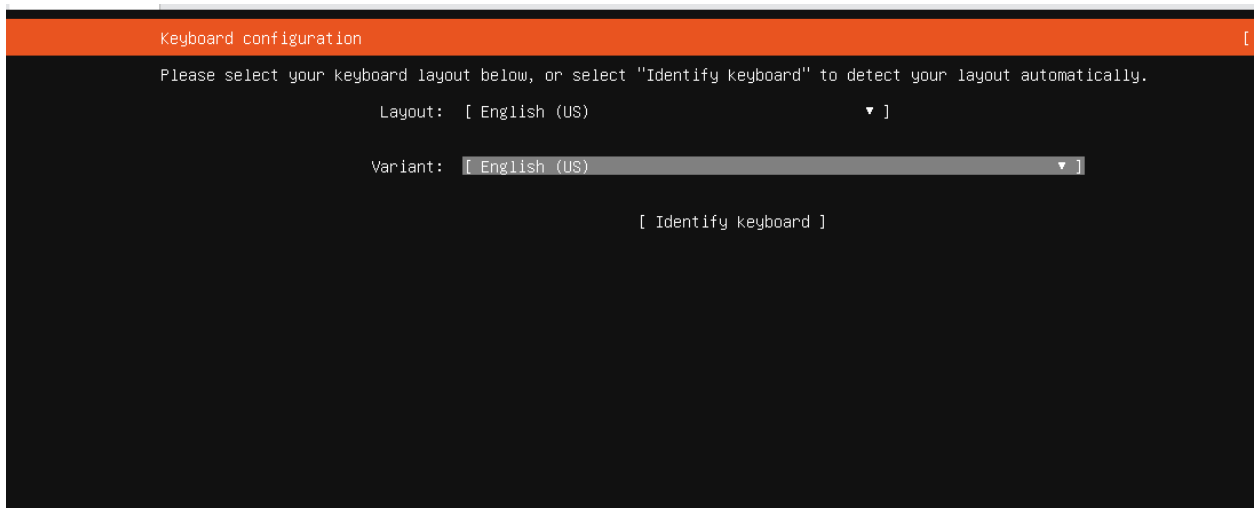


- **ubuntu_language.png** — *Shows English selected as the installation language.*

Task 6: Configure the Keyboard Layout

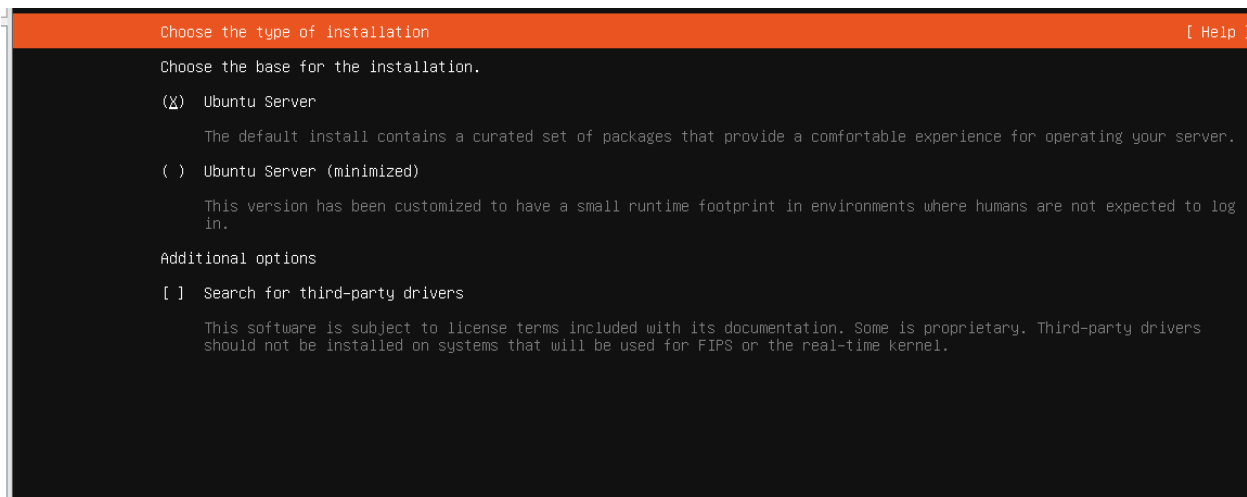


- **ubuntu_keyboard_layout.png** — *Shows English (US) selected as the keyboard layout.*



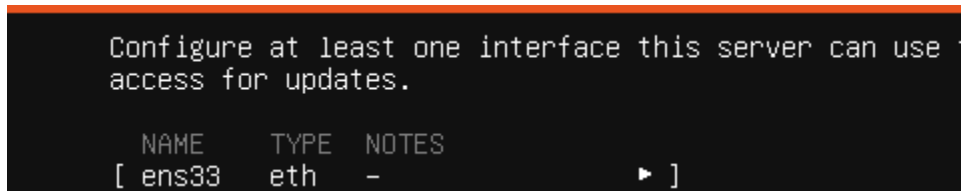
- **ubuntu_keyboard_variant.png** — *Shows English (US) selected as the keyboard variant.*

Task 7: Choose the Ubuntu Server Installation Type

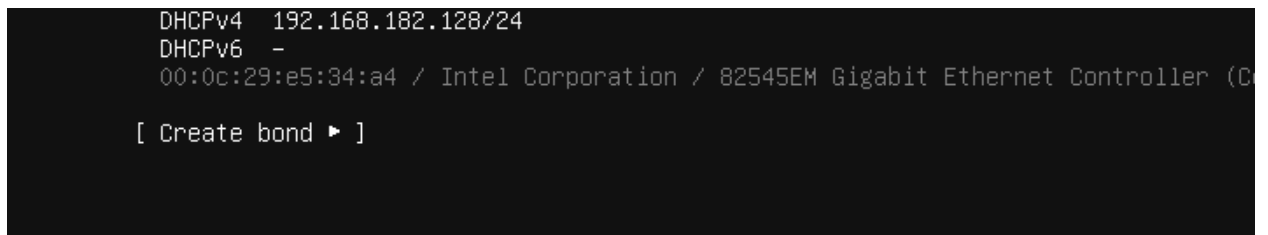


- **ubuntu_installation_type.png** — *Shows “Ubuntu Server” selected as the installation type (not MAAS).*

Task 8: Configure the Network

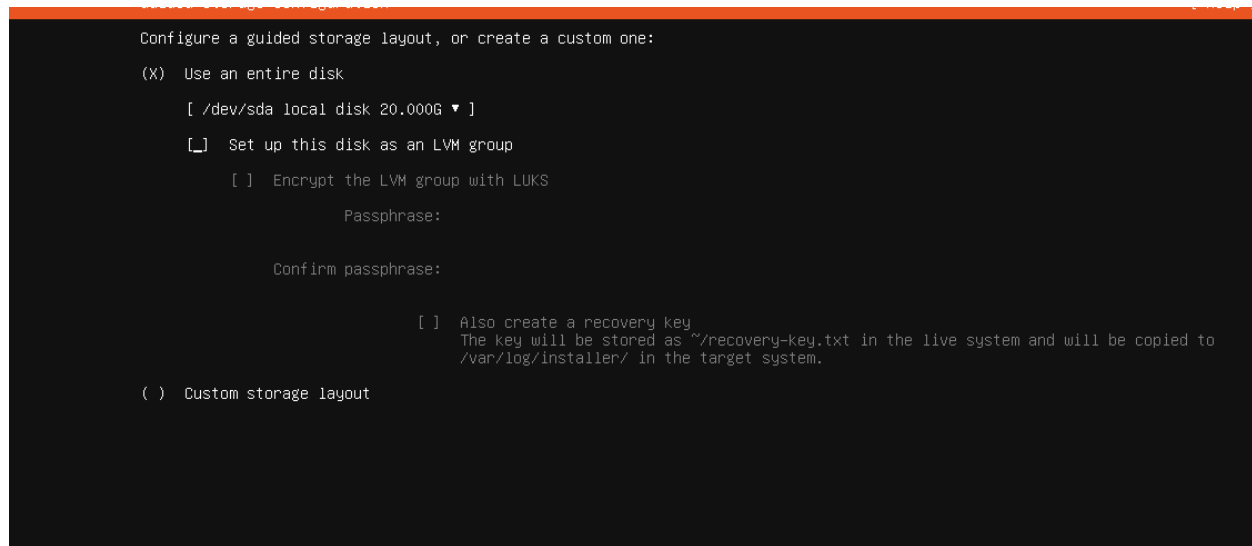


- **ubuntu_network_interface.png** — *Shows the detected network interface ens33.*

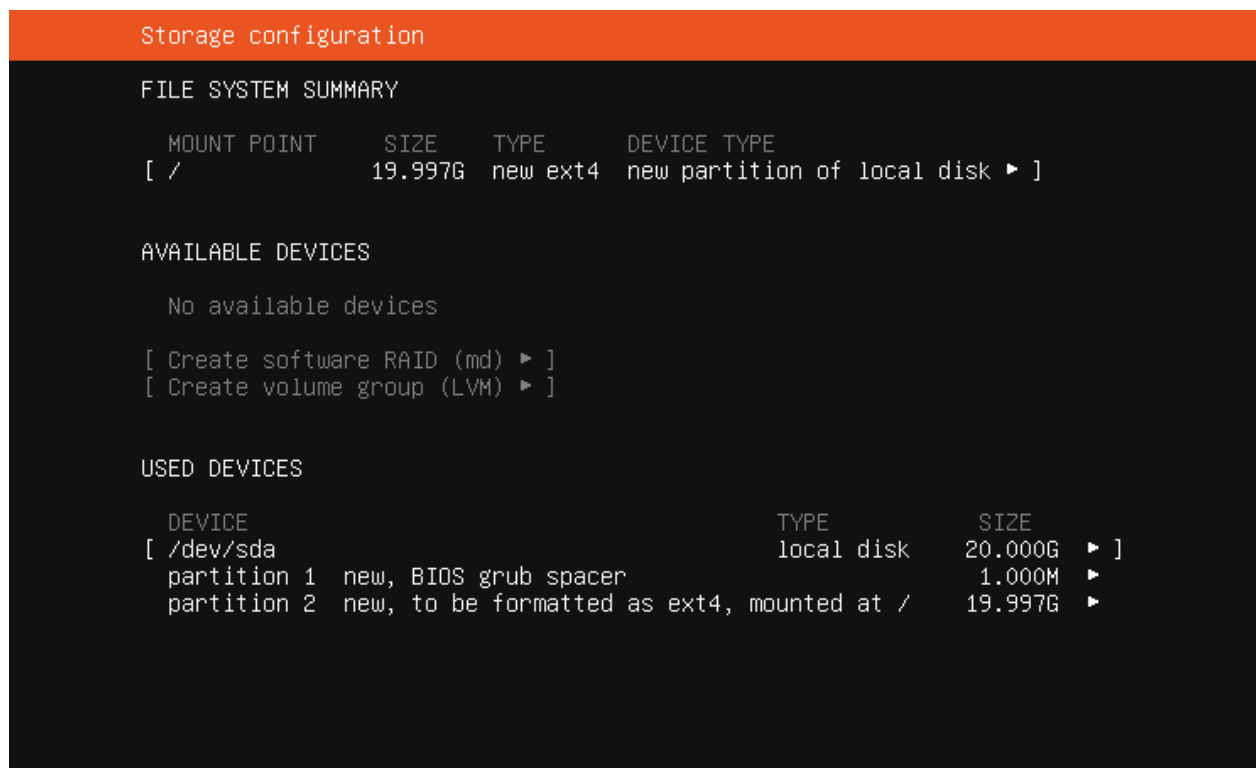


- **ubuntu_installer_network.png** — *Shows the DHCP-assigned IPv4 address 192.168.182.129/24.*

Task 9: Configure Storage



- **ubuntu_guided_storage.png** — *Shows “Use an entire disk” selected for guided storage on /dev/sda (20 GB VMware virtual disk).*



- **ubuntu_storage_configuration.png** — *Shows the proposed storage configuration using only the VMware virtual disk.*

Task 10: Select the Installation Disk

```
Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /              19.997G   new ext4   new partition of local disk ► ]

AVAILABLE DEVICES

No available devices

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

DEVICE                                TYPE      SIZE
[ /dev/sda                            local disk 20.000G ► ]
partition 1  new, BIOS grub spacer    1.000M ►
partition 2  new, to be formatted as ext4, mounted at / 19.997G ►
```

- **ubuntu_selected_disk.png** — *Shows /dev/sda selected as the installation disk. In the newer Ubuntu installer, all three of these are combined into the single "Storage configuration" screen — that's why there is only one file for them.*

```
Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /              19.997G   new ext4   new partition of local disk ► ]

AVAILABLE DEVICES

No available devices

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

DEVICE                                TYPE      SIZE
[ /dev/sda                            local disk 20.000G ► ]
partition 1  new, BIOS grub spacer    1.000M ►
partition 2  new, to be formatted as ext4, mounted at / 19.997G ►
```

-

- **ubuntu_disk_details.png** — *Shows the disk name and size: /dev/sda 20.000 GB. In the newer Ubuntu installer, all three of these are combined into the single "Storage configuration" screen — that's why there is only one file for them.*

Task 11: Review the Partitions

```
Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT      SIZE    TYPE    DEVICE TYPE
[ /              19.997G new ext4 new partition of local disk ► ]

AVAILABLE DEVICES

No available devices

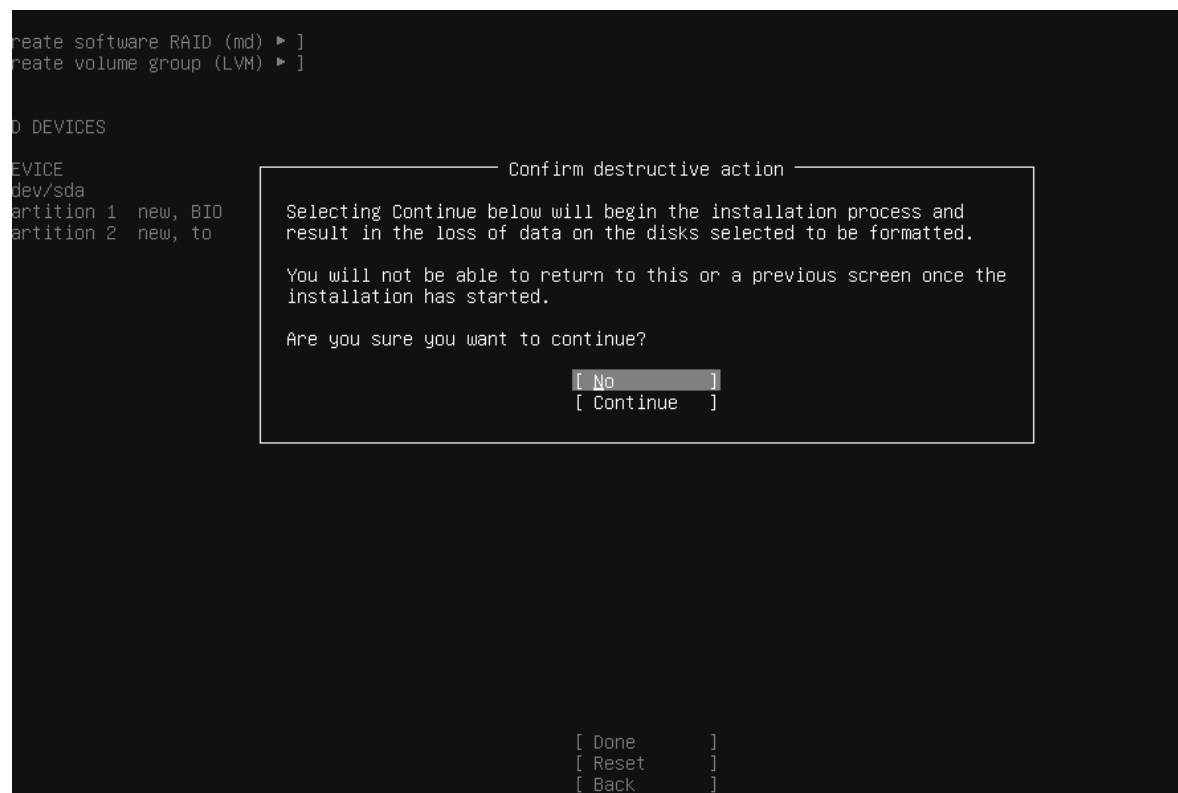
[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

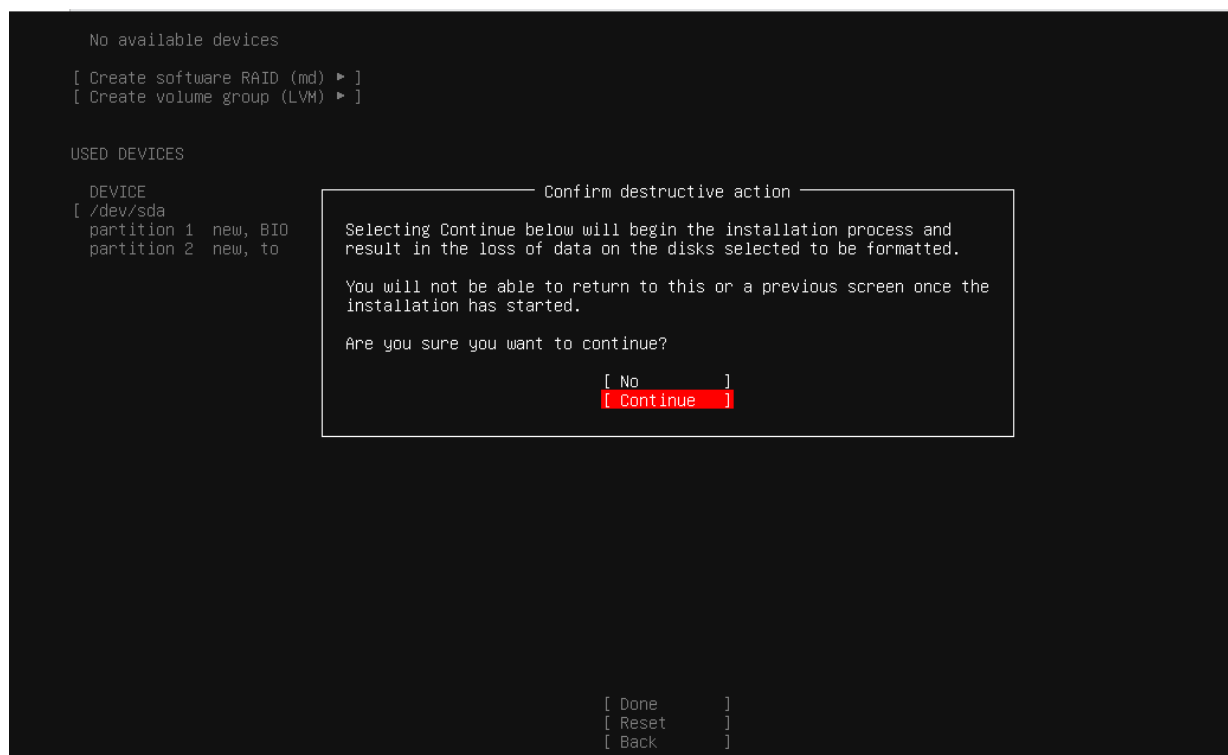
DEVICE                                TYPE    SIZE
[ /dev/sda                            local disk 20.000G ► ]
partition 1 new, BIOS grub spacer                1.000M ►
partition 2 new, to be formatted as ext4, mounted at / 19.997G ►
```

- **ubuntu_partition_layout.png** — *Shows the proposed partition layout on the VMware virtual disk before confirming. In the newer Ubuntu installer, all three of these are combined into the single "Storage configuration" screen — that's why there is only one file for them.*

Task 12: Confirm the Storage Changes



- **ubuntu_storage_warning.png** — *Shows the destructive-action warning before formatting the disk.*



- **ubuntu_storage_confirmed.png** — *Shows “Continue” selected to confirm the storage changes.*

Task 13: Configure the Ubuntu Profile

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Momina Asim

- **ubuntu_actual_name.png** — *Shows my complete actual name entered in the “Your name” field.*

Your servers name: ubuntu

The name it uses when it talks to other computers.

- **ubuntu_server_name.png** — *Shows ubuntu entered as the server name.*

Pick a username: mominaasim-dev

- **ubuntu_username.png** — *Shows mominaasim-dev (my GitHub username) entered as the Ubuntu username.*

Profile configuration
[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Momina Asim

Your servers name: ubuntu
The name it uses when it talks to other computers.

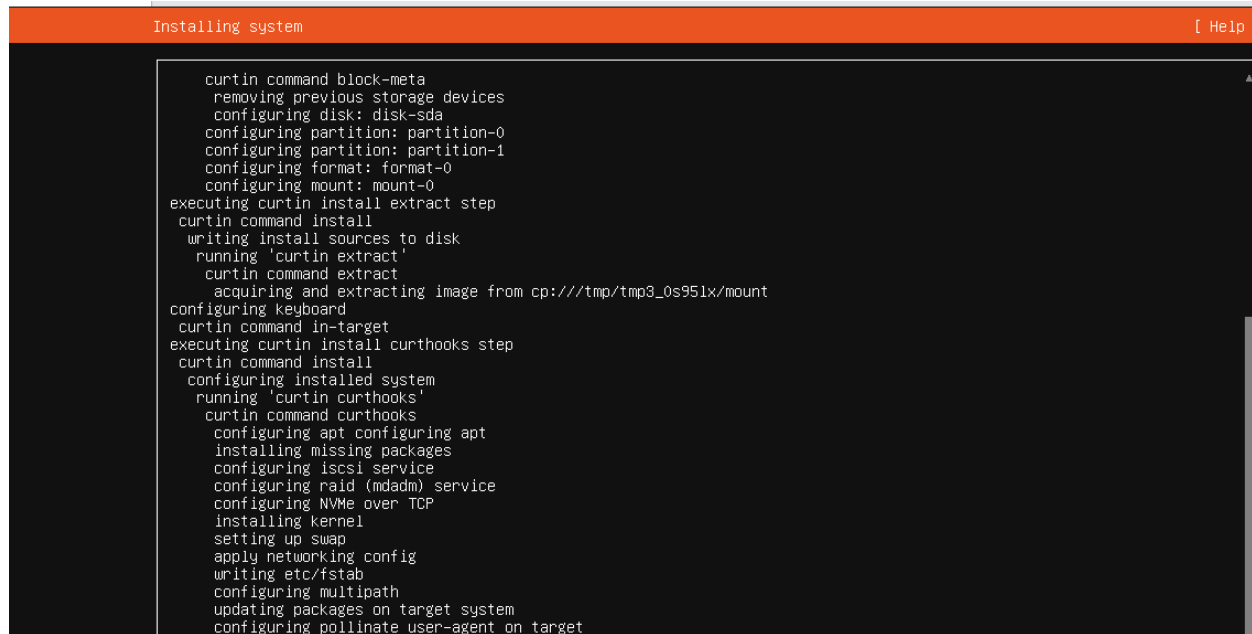
Pick a username: mominaasim-dev

Choose a password:

Confirm your password:

- **ubuntu_profile_setup.png** — *Shows the completed profile page with all fields filled (password hidden).*

Task 14: Complete the Software Installation

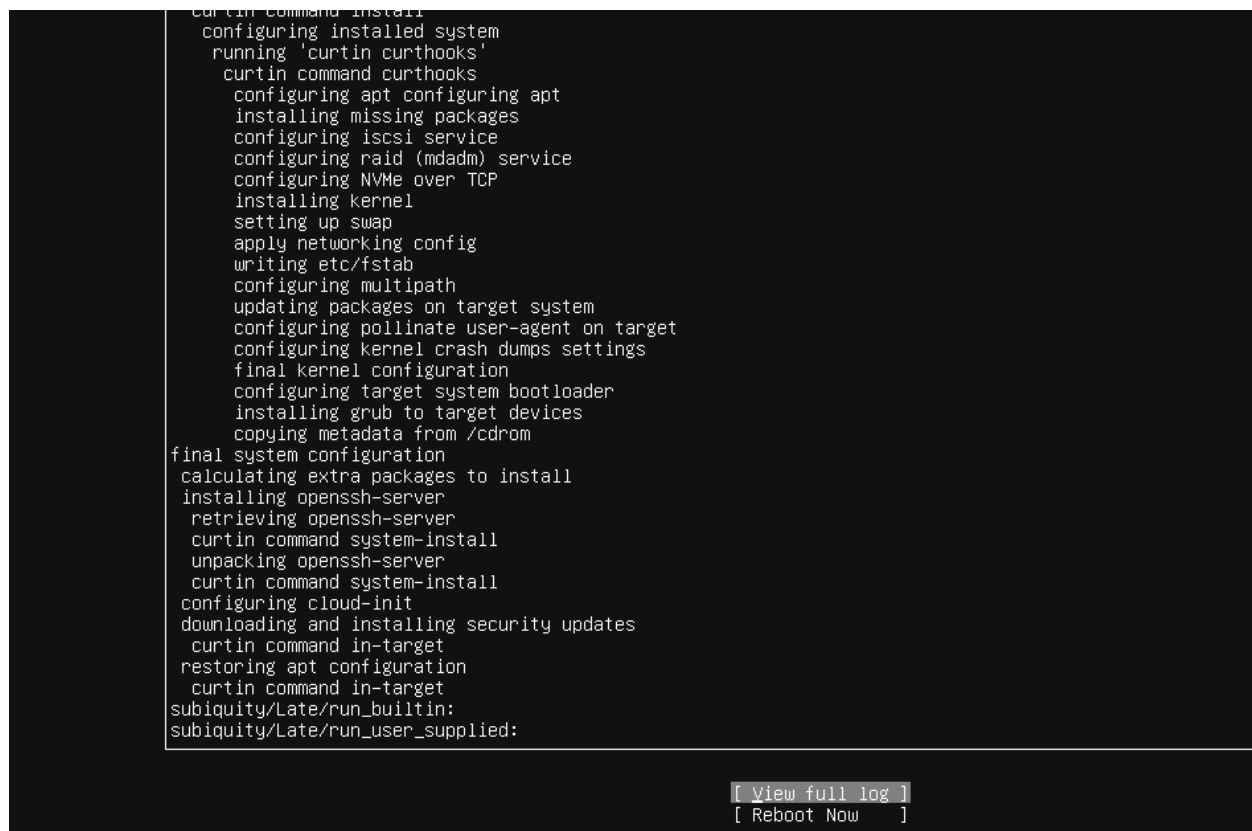


```
Installing system [ Help ]

curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring format: format-0
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp3_0s951x/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt
    configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
```

- **ubuntu_installation_progress.png** — *Shows the Ubuntu Server installation in progress.*

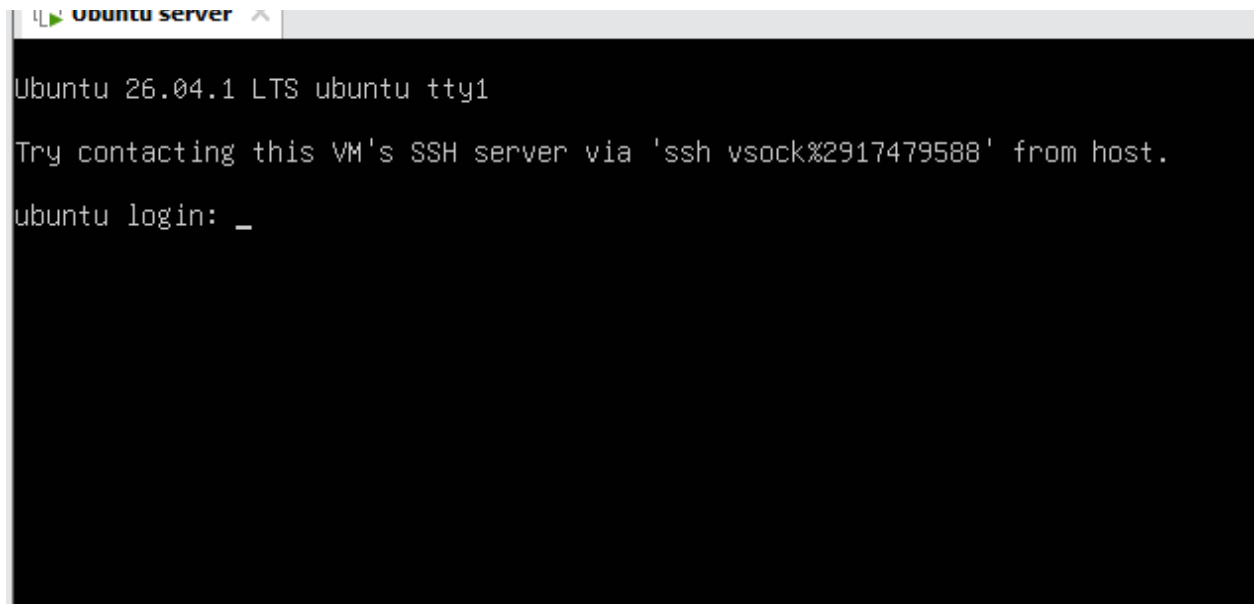
Task 15: Reboot and Verify the Ubuntu Identity



```
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt
    configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
  configuring cloud-init
  downloading and installing security updates
  curtin command in-target
  restoring apt configuration
  curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

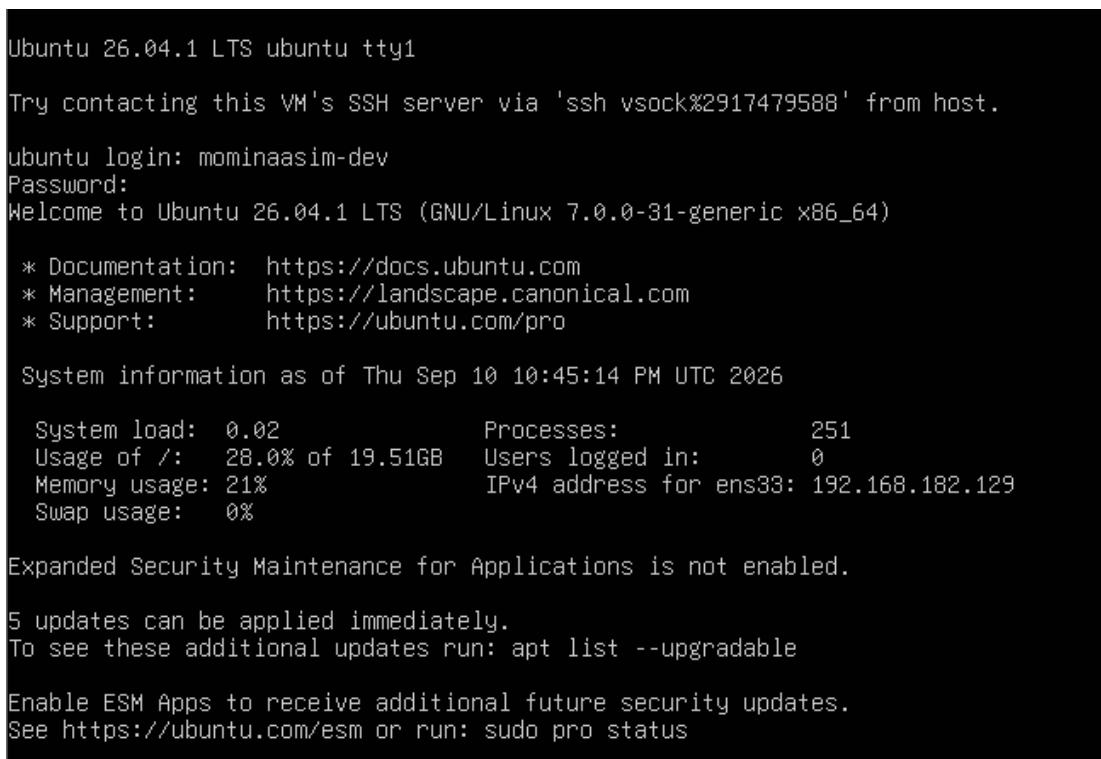
[View full log]
[Reboot Now]

- **ubuntu_installation_complete.png** — *Shows the “Installation complete!” screen with the “Reboot Now” option.*



```
Ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock%2917479588' from host.
ubuntu login: _
```

- **ubuntu_login_screen.png** — *Shows the Ubuntu Server login prompt after reboot.*



```
Ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock%2917479588' from host.
ubuntu login: mominaasim-dev
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 10 10:45:14 PM UTC 2026

System load:  0.02                Processes:            251
Usage of /:   28.0% of 19.51GB    Users logged in:     0
Memory usage: 21%                IPv4 address for ens33: 192.168.182.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

- **ubuntu_terminal_login.png** — *Shows successful login to the Ubuntu Server terminal.*

```

mominaasim-dev@ubuntu:~$
mominaasim-dev@ubuntu:~$

```

- **ubuntu_identity_verified.png** — *Shows the terminal prompt mominaasim-dev@ubuntu:~\$ verifying the identity.*

Task 16: Find the Ubuntu Server IP Address

```

mominaasim-dev@ubuntu:~$
mominaasim-dev@ubuntu:~$ [ 487.296573] e1000 0000:02:01:0 ens33: Reset adapter
ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:e5:34:a4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29e534a4
    inet 192.168.182.129/24 metric 100 brd 192.168.182.255 scope global dynamic ens33
        valid_lft 1682sec preferred_lft 1682sec
    inet6 fe80::20c:29ff:fee5:34a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

- **ubuntu_ip_addr_command.png** — *Shows the ip addr command and its output.*

```

    altname enx000c29e534a4
    inet 192.168.182.129/24 metric 100 brd 192.168.182.255 scope global dynamic ens33
        valid_lft 1682sec preferred_lft 1682sec
    inet6 fe80::20c:29ff:fee5:34a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mominaasim-dev@ubuntu:~$ sudo systemctl status ssh

```

- **ubuntu_ip_address.png** — *Shows the IPv4 address 192.168.182.129 for interface ens33 (not 127.0.0.1).*

Task 17: Connect to Ubuntu Server from Windows

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>ssh mominaasim-dev@192.168.182.129
The authenticity of host '192.168.182.129 (192.168.182.129)' can't be established.
ED25519 key fingerprint is SHA256:Hb7nYASpgH5b6f2noj70mbdX6rKtr+P6liDbLdi7Yt4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

- windows_ssh_command.png — *Shows the SSH command ssh mominaasim-dev@192.168.182.129 executed from Windows.*

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.182.129' (ED25519) to the list of known hosts.
mominaasim-dev@192.168.182.129's password:
```

- windows_ssh_fingerprint.png — *Shows yes typed to accept the server fingerprint.*

```
C:\Users\hp>ssh mominaasim-dev@192.168.182.129
mominaasim-dev@192.168.182.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 11 03:20:14 AM UTC 2026

System load:  0.15           Processes:            221
Usage of /:   28.1% of 19.51GB Users logged in:      0
Memory usage: 24%           IPv4 address for ens33: 192.168.182.129
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

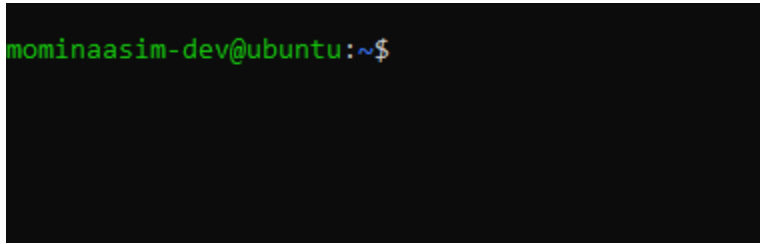
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

- windows_ssh_login.png — *Shows the successful SSH login to Ubuntu Server.*



- windows_ssh_identity.png — *Shows the remote SSH prompt mominaasim-dev@ubuntu:~\$ confirming identity.*

Task 18: Prepare Lab1 Solution PDF

Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Momina Asim



Registration number: fa24b1-se-069

Course code: Cloud Computing

Section: A|

Github username: mominaasim-dev

- solution_title_page.png — *Shows the completed title page with lab number, name, registration number, course code, section, and GitHub username.*

- **lab1_solution_pdf.png** — *Shows the exported Lab1_Solution.pdf file saved at Labs/Lab01/.*